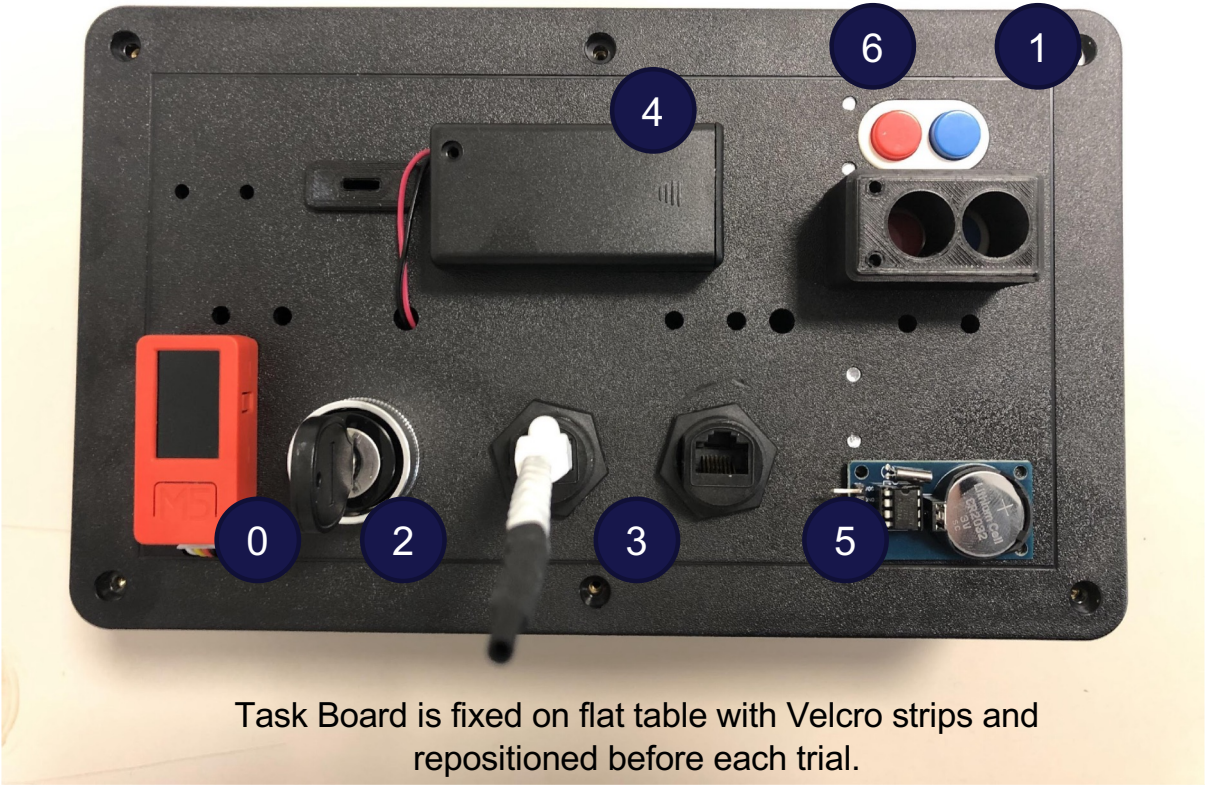
Robothon® – Task board & Trial Protocol

0. Press M5 Button to Start Trial
1. Find Board & Press Button
2. Get Key & Activate Key Switch
3. Pick & Move Ethernet Plug
4. Open & Extract AA Batteries
5. Extract Coin Battery
6. Press Stop Button to End Trial

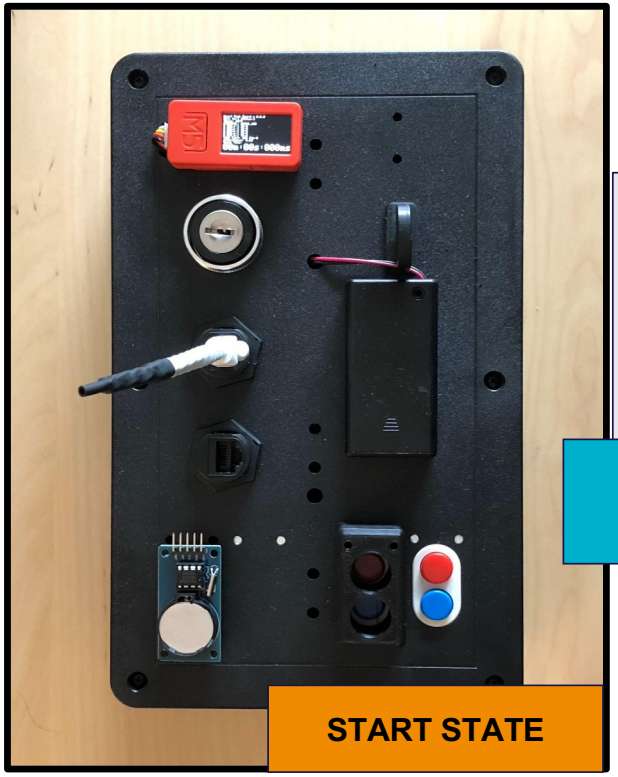
Trial Time Starts!

Trial Time Ends!

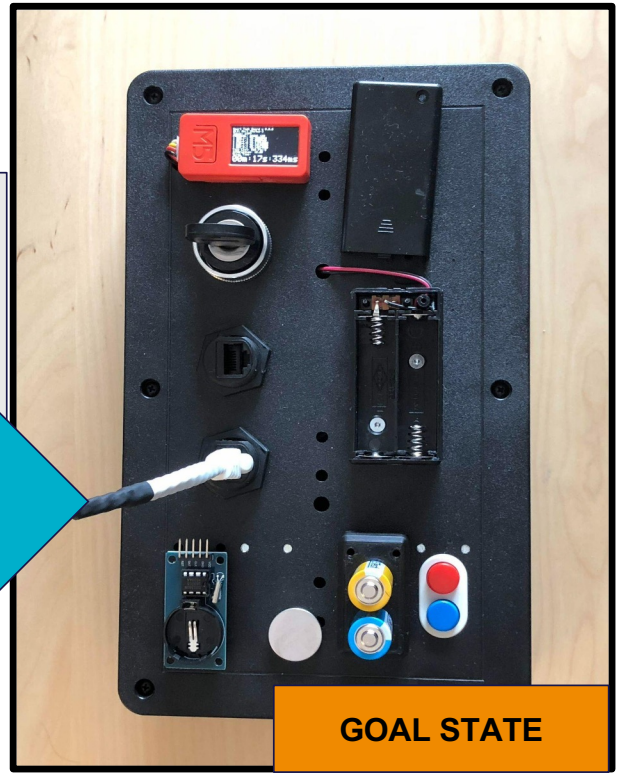


Task Board is fixed on flat table with Velcro strips and repositioned before each trial.

Task Board Tasks: Start and Goal States



Robot Platform
10min.
No assistance.



Competition Rules

- During the development period, each team must develop their robot platform to move the task board from its starting state to the goal state without any human intervention.
- Prior to starting a trial, your team will be asked to pick up and replace it on the velcro strips to randomize the board's position between trials to prevent pre-programmed solutions.
- Points are awarded to your team for each progressive step automatically after their robot platform achieves when the task board is connected to the internet.
- Teams who earn an equal number of points will be distinguished by competition time (faster is better).



Don't forget to practice resetting your board before each trial!

	Score Category	Description	Rubric	Max Value
Submission Package	Recording	Continuous video recording (max length 10min) with points awarded from task board Bonus for Robustness.	Task 1: 1pt, Task 2: 4pts, Task 3: 4pts Task 4: 6pts, Task 5: 5pts Max: 20pts •Bonus multiplier for back-to-back demonstrations (task board must be repositioned between trials).	20 x 5 = 100
	Solution Transferability Demo	Team includes video demonstration of battery extraction of a device of their own choosing, e.g. smoke detector or TV remote.	•Local E-Waste Assignment (graded on 1-10 scale, averaged & mapped). Graded by jury for authenticity, number of fixtures required, task success, task execution time.	30
	Documentation	Team provides accurate description of their robot platform along with instructions on how to start their system.	•Repository has: - equipment used list, 10pts - software dependency list, 10pts - quick start guide. 10pts	30
Live Demo	Live Presentation	Team demonstrates solution to task board during 30min video call over Zoom. Bonus for Flexibility	•Team Interview - 1st Trial Demo, 20pts - 2nd Trial Demo, 20pts (Jury's Sequence) •Bonus awarded if team completes 2nd attempt with task order specified by jury in session.	20 + 20 = 40
	Total Team Score	-	-	**160 + 40 = 200**

Team Evaluation Scorecard Design

Team Name:				Date:	Time:
Robot Platform	Manipulator:	End Effector:		Sensor1: Sensor2: Sensor3:	

Task Board Obstacles

Task	**Localize Board**		**Task**	**Move Plug**		**Task**	**Key Insertion**		**Task**	**AA Batt Removal**		**Task**	**Coin Batt Removal**	
Level 1	Points		Level 2	Points		Level 3	Points		Level 4	Points		Level 5	Points	
Subtask	Avail.	Earned	Subtask	Avail.	Earned	Subtask	Avail.	Earned	Subtask	Avail.	Earned	Subtask	Avail.	Earned
Press Blue Button	1		Pick up Ethernet Plug	1		Pick up Key	1		Remove battery lid	2		Remove coin batt	4	
			Insert Ethernet Plug	3		Insert and Turn Key	3		Remove battery (2pts each)	4		Coin batt placed on plate	1	
Task 1 Subtotal			Task 2 Subtotal			Task 3 Subtotal			Task 4 Subtotal			Task 5 Subtotal		
Max Pts	1		Max Pts	4		Max Pts	4		Max Pts	6		Max Pts	5	
Method Used:			Method Used:			Method Used:			Method Used:			Method Used:		

Transferability Concept Object:	Presentation Score for Transferability (1-10 with 10 as highest):

Team Evaluations: 2 Parts

Part I:
Submission Package Due May 29, 2022

Github Repository
Web Link

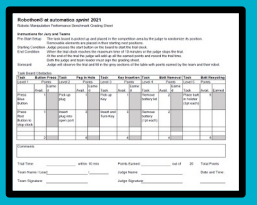
Video Presentation (5min max)
.mp4

Uncut Demo Recording (10min max)
.mp4

Send 3 links in a **single PDF** via email to info@robothon-grand-challenge.com by midnight CEST on 29 May 2022.

Part II:
Live Demo to Jury May 30-June 2





Scorecard



Web Dashboard